

Seat No.	
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Set **P**

**B.Sc. (E.C.S.) (Semester - I) (New) (NEP CBCS) Examination:
October/November – 2025
Digital Electronics and Microprocessors (G06-0103)**

Day & Date: Thursday, 13-11-2025
Time: 12:00 PM To 01:30 PM

Max. Marks: 30

- Instructions:** 1) All questions are compulsory.
2) Draw neat diagrams and give equations wherever necessary.
3) Figures to the right indicate full marks.
4) Use of logarithmic table and calculator is allowed.
(At. Wts.: H=1, C=12, O=16, N=14, Na=23, Cl=35.5)

Q.1 Multiple Choice Questions:**06**

- 1) _____ basic gates.

a) nand	b) nor
c) not	d) all
- 2) Capacity of flip flop to store _____ bit.

a) 1	b) 2
c) 3	d) 4
- 3) HA consist _____ i/p and _____ o/p.

a) 2,2	b) 3,2
c) 3,3	d) 1,1
- 4) Shift register consist of _____.

a) all	b) sr ffs
c) jk ffs	d) d ffs
- 5) Data bus of 8085 is _____ bit.

a) All	b) 14
c) 8	d) 16
- 6) _____ Data transfer instruction.

a) push	b) add
c) inc	d) none

Q.2 Answer the following questions. (Any three)**06**

- a) Define Full Adder.
- b) Define Encoder.
- c) Define Asynchronous Counter.
- d) Define Shift Register.
- e) Define Addressing mode.

- Q.3 Write Shorts Notes. (Any two)** **06**
- a) Application of logic gates.
 - b) Application of mux.
 - c) Master-slave FF's.
- Q.4 Answer the following questions. (Any two)** **06**
- a) Explain SISO shift register.
 - b) Write feature of 8085.
 - c) explain 4:1 MUX.
- Q.5 Answer the following question. (Any one)** **06**
- a) Draw symbol of and truth table of NAND, NOR logic gate.
 - b) Define and explain addressing mode with one example of 8085.